



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. (ELECTRONICS AND COMMUNICATION ENGINEERING)

EXAMINATIONS - JAN/FEB 2023

**SUBJECT : ECE 2152 - COMPUTER ORGANIZATION AND ARCHITECTURE
(MAKEUP)**

COMPUTER ORGANIZATION AND ARCHITECTURE [ECE 2152]

Marks: 50

Duration: 180 mins.

A

Answer all the questions.

1A) For the expression $X = (A-B/C)/(D + E/F + G*H)$, write the program using 3-address, 2-address, 1-address and 0-address instructions. Assume source and destination operands are memory locations. (5)

1B) In a computer instruction format, the instruction length and size of the address field are 12 bits and 4 bits respectively. The instruction set has 14 two addresses and 50 zero address instructions. Using expanding opcode technique, compute the number of one address instructions that can be added to the instruction set. Show the detailed schematic of the instruction format. (3)

1C) With the help of a block diagram explain the working of general register based processor. (2)

2A) By showing all the steps multiply 14 and -5 using Booth's algorithm. Verify the result using Modified Booth's algorithm. (5)

2B) Design and implement a digital circuit using parallel adder and 2:1 MUX that will work as follows. Assume A is 4-bit data:

S1	S0	Function F
0	0	Shift left (A)
0	1	$(1\ 1\ 1\ 1)_2$
1	0	Shift left (A) plus 1
1	1	$(0\ 0\ 0\ 0)_2$

(3)

2C) Assume the CSA add time is half of the CPA time. Calculate the total time required to add, by giving suitable expressions in a fully parallel 16 operand summation. Find the number of guard bits required. If a Wallace tree structure is used, what is the expression for total time taken? (2)

3A) Design a microprogrammed control unit for Booth's multiplier by giving the RTL code. Explain the control memory organization of Booth's multiplier with a neat diagram. (5)

3B) Build the hardware to implement each of the following register transfers:
i) If $Q[1]=0$ AND $Q[0]=1$ then $A \leftarrow A+1$ else $A \leftarrow A-1$
ii) If $Q[1:0]=01$ OR $Q[1:0]=10$ then $A \leftarrow B$
Assume A, B and Q are 4-bit registers. (3)

3C) Convert the following decimal numbers to 32-bit IEEE floating point representation. Write the result in hexadecimal format. (2)

- a) 45.125
- b) 0.0125

- 4A) Distinguish the difference between maskable and non-maskable interrupts. Explain with neat diagrams, software polled approach for connecting multiple interrupts to a single interrupt line. List its advantages and disadvantages. (5)
- 4B) A computer system needs a 1K X 8 RAM. Assuming 256 X 4 RAM chips are available, show how to build required RAM. Draw a neat logic diagram of your implementation. (3)
- 4C) A certain memory system has a 32MB main memory and a 64 KB Cache. Block are 16 bytes in sizes. Show the main memory address format to determine the length of TAG, Cache line number and WORD fields for the Direct mapping technique. (2)
- 5A) Explain in detail the characteristic features of VLIW architecture. Mention its advantages and disadvantages. (5)
- 5B) A processor has to execute the following operation $R0 \leftarrow R2 + R1$. Calculate the number of clock cycles required in each of the following cases (a) single bus RALU (b) two bus RALU and (c) three bus RALU. Also justify the answer. (3)
- 5C) With neat diagram discuss the timer/counter operation in 8051 microcontrollers. (2)

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